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- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

# Bayesian Statistics and Data Analysis
# Course information

Måns Magnusson
Department of Statistics, Uppsala University

# Who am I?

1. Associate professor in Statistics
2. Teaching: Machine Learning and Bayesian Statistics
3. Research: Machine Learning and Bayesian Statistics
    3.1 Bayesian Statistics and computational statistics
    3.2 Data-centric machine learning
    3.3 Statistical inference from textual data
    3.4 Applications: Law, sociology, political science, and medicine

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# Section 1

# Course information

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The aims of this course are that, after this course you should:

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# Course information

The aims of this course are that, after this course you should:

1. have knowledge in basic concepts, philosophy, and perspectives in Bayesian Statistics,
2. derive posterior distributions in simple situations,
3. derive and use predictive distributions,
4. identify and formulate Bayesian probabilistic models for analysis and predictions,
5. estimate models using contemporary computer-based methods for posterior approximations,
6. understand and use basic principles for decisions under uncertainty,
7. have knowledge about and be able to use Bayesian methods for model comparisons,
8. be able to critically evaluate Bayesian methods,
9. report, orally and in writing, a Bayesian statistical analysis

# Pre-requisites

- Basic probability theory
  - probability, probability density, distribution
  - sum, product rule, and Bayes' rule
  - expectation, mean, variance, median

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# Pre-requisites

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  - probability, probability density, distribution
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  - expectation, mean, variance, median
- Basic linear algebra and calculus
- Basic visualisation techniques (R or Python)
  - histogram, density plot, scatter plot
- *Note!* This is a masters course in Statistics.

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- *Note!* This is a masters course in Statistics.

*First assignment is a recap.*

# Who are you? (very brief)



1. Name
2. What is your expectation on this course? Why did you choose it?

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# Course Outline

Main components:

- Core Content (9 lecture blocks)

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Main components:

- Core Content (9 lecture blocks)
- Assignments (8 individual assignments)
- Oral exams connected to the assignments
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Exact dates and details; see the course page.

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# Core Content

- Every week: lectures (approx. 2-4h)
    - Online video material and reading assignments (approx. 2-4h, 50-90 pages a week)
    - Lecture(s): present overall theory and content (overview)
    - Assignment(s): Computational and theoretical individual work. (approx. 12-16h). Deadline Sundays 23.59.
      *Start monday morning every week!*

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- Recommended workflow for each week
  - Do the reading assignments
  - Watch the videos (although, optional)
  - Do self-study exercises
  - Start with the assignment
  - Attend lecture (bring questions!)
  - Attend Zoom datalabs (bring questions! Helps with debugging Stan code)
  - Submit the assignment
  - Review the oral exam questions and practice explaining answers in groups

# Section 2

# Assignments

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# Assignments

1. Core components and concepts and state-of-the-art methods

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   3.1 The week of the assignment
   3.2 The last day of the course
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- Assignment PDFs may contain hidden machine-readable integrity text. Do not copy or upload the full assignment text into an LLM.

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# Oral exams

- Three oral exams during the course:
    1. Assignments 1–3
    2. Assignments 4–6
    3. Assignments 7–8

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- If the first answer is not sufficient, a second question from the same assignment is drawn.
- Passing the oral exam is required to receive VG for the corresponding assignment.
- Each oral exam has three attempts in total: one ordinary opportunity and two re-examination opportunities.

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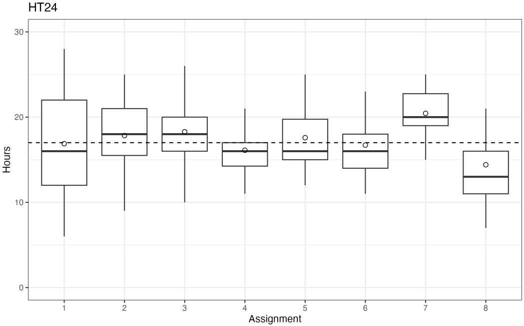
- We also supply a knitR LaTeX template suitable for Overleaf.

# Course Workload

- Conclusions from 2024
  - Slightly too large workload in assignment 7




Student course workload

# Stan

- Stan is a probabilistic programming framework (PPF) and ecosystem
- 40+ developers, 100+ contributors, 100K+ users
- R, Python, Julia, Scala, Stata, Matlab, command line interfaces
- More than 120 R packages using Stan
- Many packages to support diagnostics and workflow
- Can be used for frequentist inference as well
- Alternative PPF exists, Turing (Julia), Pyro (PyTorch), etc.

mc-stan.org

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# Section 3

## Mini-project

# Mini-project

- See project instructions on webpage for details.
- Two possible tracks: ordinary data analysis on real data, or paper replication.
- 2-3 students.

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- Supply a one-page project proposal in the middle of the course.

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- Two possible tracks: ordinary data analysis on real data, or paper replication.
- 2-3 students.
- Supply a one-page project proposal in the middle of the course.
- Ideally, use a model not presented in this course.
- Project will last two weeks (half time) - but start earlier.
- The project should use Stan.
- Approximate 40 hours of work *per student*.

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- The project should result in a 4 page report (PDF) using the ICML LaTeX template (see course page).
- Project oral presentation (10-15 minutes)

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- The project should use Stan.
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- The project should result in a 4 page report (PDF) using the ICML LaTeX template (see course page).
- Project oral presentation (10-15 minutes)
- The first author is corresponding author

# Mini-project

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- 2-3 students.
- Supply a one-page project proposal in the middle of the course.
- Ideally, use a model not presented in this course.
- Project will last two weeks (half time) - but start earlier.
- The project should use Stan.
- Approximate 40 hours of work *per student*.
- The project should result in a 4 page report (PDF) using the ICML LaTeX template (see course page).
- Project oral presentation (10-15 minutes)
- The first author is corresponding author
- *This year*:
    - Paper replication track: open paper + Stan implementation + reproduce main result + prepare for `posteriordb`.
    - Mini-project discussions: after lectures and at Thursday Zoom sessions (first from next week).

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# Master theses using Stan

- New thing this year: Joint supervision and work with researchers and students (PhD and masters)
- For students that want to work closely to research(ers)
- Headed by me and Sara Hamis at IT
- First step: a joint hackathon in mid november (morning + lunch).

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- Six project (three from stats and three from IT) on Stan and Bayes:
- At Statistics:
    - Modelling of pollen data in Ancient greece (with Anton Bonnier)
    - Using the Gumbel Softmax to do missing data imputation with discrete parameters (with Jakob Torgander)
    - Modelling proteins in pharmacometrics

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# Section 4

# Practicalities

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# Practicalities

- Course page/content: Github – please do a PR or submit an issue if something is wrong!
  https://github.com/MansMeg/BSDA/issues

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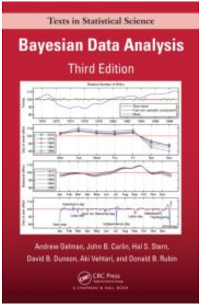
- Acknowledgements: Aki Vehtari
- Teaching assistant: Väinö Yrjänäinen

# Literature

- Book: Gelman, Carlin, Stern, Dunson, Vehtari & Rubin: Bayesian Data Analysis, Third Edition. (online pdf available)




- Additional articles and blog posts (see reading list per week)

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# Section 5

# Examination

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2. To pass with distinction (VG): 7/10 VG points
3. If everything is correct in an assignment (¿90%), 1 VG point is awarded *on the first submission deadline*.
4. Assignment VG points require passing the oral exam for that assignment.
5. The mini-project is worth 2 VG-points (if it is passed with distinction).
6. Ph.D. students: I suggest you get VG to pass the course. Make the project a potential paper.
7. Reassesement of grades (supply form to course admin)
8. Failing the course: You will need to redo all assignments and mini-project next year.
9. LLMs may be used for learning, explanations, and feedback, but not to produce submitted solutions.

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# Section 6

# Course improvements

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# Course improvements since last time

- Better balance between Assignment 7 and 8.
- Hidden integrity text and Large Language Model use.
- New idea (test) of mini project.

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# Section 6

## Course improvements

# Comments from previous students

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- Attend the lectures and start with the assignments early. Because they are really fun if you get passed the time pressure aspect
- Start the assignments as soon as possible. It is perfectly reasonable to be done with the assignments before the weekend, so aim for that.
- Do the course properly, it is worth the effort

# Questions?

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Questions?